Algebraic Geometry 8
Lecture notes with examples

July 29, 2026

Contents

Solutions: Generic Points, Dimension, Codimension, and Counterexamples

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1 Problem 1: Generic Point of a Closed Irreducible Subset

Statement

Let X be a scheme and let $Z \subseteq X$ be a closed irreducible subset. Show that there exists a unique point $\eta \in Z$ such that
$$\overline{\{\eta\}} = Z.$$

Solution

We prove existence first.

Choose a point $x \in Z$. Since X is a scheme, there exists an affine open neighbourhood
$$U = \operatorname{Spec} A \subseteq X$$
such that $x \in U$.

Then
$$Z \cap U$$
is nonempty, because it contains x. It is also open in Z, since U is open in X, and it is closed in U, since Z is closed in X.

Because Z is irreducible and $Z \cap U$ is a nonempty open subset of Z, the subset $Z \cap U$ is irreducible.

Since $U = \operatorname{Spec} A$, every closed subset of U has the form
$$V(I)$$
for some ideal $I \subseteq A$. The irreducible closed subsets of $\operatorname{Spec} A$ are exactly the sets of the form
$$V(\mathfrak{p}),$$
where $\mathfrak{p} \subseteq A$ is a prime ideal.

Therefore, since $Z \cap U$ is closed and irreducible in U, there exists a unique prime ideal $\mathfrak{p} \subseteq A$ such that
$$Z \cap U = V(\mathfrak{p}).$$

Let
$$\eta = \mathfrak{p} \in \operatorname{Spec} A = U.$$

Then in U, we have
$$\overline{\{\eta\}}^U = V(\mathfrak{p}) = Z \cap U.$$

Now let
$$W = \overline{\{\eta\}}^X$$
be the closure of η in X. Since $\eta \in Z$ and Z is closed in X, we get
$$W \subseteq Z.$$

Moreover, since U is an open subset containing η, closure commutes with restriction to U:

$$W \cap U = \overline{\{\eta\}}^X \cap U = \overline{\{\eta\}}^U.$$

Hence

$$W \cap U = Z \cap U.$$

We now show that $W = Z$. Suppose, for contradiction, that

$$W \neq Z.$$

Then $Z \setminus W$ is a nonempty open subset of Z. Also, $Z \cap U$ is a nonempty open subset of Z. But since

$$Z \cap U = W \cap U,$$

we have

$$Z \cap U \subseteq W.$$

Therefore

$$(Z \setminus W) \cap (Z \cap U) = \varnothing.$$

Thus Z contains two disjoint nonempty open subsets, namely $Z \setminus W$ and $Z \cap U$. This contradicts the irreducibility of Z.

Therefore

$$W = Z.$$

Hence

$$\overline{\{\eta\}} = Z.$$

So existence is proven.

It remains to prove uniqueness.

Suppose $\eta, \eta' \in Z$ satisfy

$$\overline{\{\eta\}} = Z$$

and

$$\overline{\{\eta'\}} = Z.$$

Then

$$\overline{\{\eta\}} = \overline{\{\eta'\}}.$$

Every scheme is a T_0-topological space. In a T_0-space, two points with the same closure are equal. Therefore

$$\eta = \eta'.$$

Hence there exists a unique point $\eta \in Z$ such that

$$\boxed{\overline{\{\eta\}} = Z.}$$

This point is called the **generic point** of Z.

2 Problem 2: Dimension and Codimension for Varieties

Statement

Let X be a variety over $\operatorname{Spec} k$. Show:

(a) For any closed point $p \in X$,

$$\dim X = \dim \mathcal{O}_{X,p}.$$

(b) If $Y \subseteq X$ is a closed subset of X, define

$$\operatorname{codim}(Y, X) = \inf_{Z \subseteq Y} \operatorname{codim}(Z, X),$$

where the infimum is taken over all closed irreducible subsets Z of Y. Show that

$$\operatorname{codim}(Y, X) = \inf\{\dim \mathcal{O}_{X,y} \mid y \in Y\}.$$

(c) If $Y \subseteq X$ is closed, then

$$\dim Y + \operatorname{codim}(Y, X) = \dim X.$$

(d) If $U \subseteq X$ is a nonempty open subset, then

$$\dim U = \dim X.$$

(e) $\dim X$ coincides with the transcendence degree of $K(X)$ over k:

$$\dim X = \operatorname{trdeg}_k K(X).$$

We may use the following facts from commutative algebra.
If B is a finitely generated k-algebra and a domain, and if $\mathfrak{p} \subseteq B$ is a prime ideal, then

$$\operatorname{height}(\mathfrak{p}) + \dim B/\mathfrak{p} = \dim B.$$

If K is the field of fractions of B, then

$$\dim B = \operatorname{trdeg}_k K.$$

General Setup

Since X is a variety over k, we assume that X is an integral scheme of finite type over k. Let

$$K(X)$$

denote the function field of X.
If

$$U = \operatorname{Spec} B \subseteq X$$

is a nonempty affine open subset, then B is a finitely generated k-algebra and a domain. Moreover,

$$\mathrm{Frac}(B) = K(X).$$

By the given commutative algebra fact,

$$\dim B = \mathrm{trdeg}_k \mathrm{Frac}(B).$$

Since $\mathrm{Frac}(B) = K(X)$, this gives

$$\dim B = \mathrm{trdeg}_k K(X).$$

This observation will be used several times.

2.1 Part (a): Closed Points and Local Dimension

Let $p \in X$ be a closed point. We need to prove that

$$\dim X = \dim \mathcal{O}_{X,p}.$$

Choose an affine open neighbourhood

$$U = \mathrm{Spec}\, B \subseteq X$$

of p. The point p corresponds to a prime ideal $\mathfrak{m} \subseteq B$. Since p is closed in X, and hence closed in the affine open neighbourhood U, the ideal $\mathfrak{m}$ is maximal.

Thus

$$p \leftrightarrow \mathfrak{m} \in \mathrm{Spec}\, B.$$

Because B is a finitely generated k-algebra and $\mathfrak{m}$ is maximal, the quotient $B/\mathfrak{m}$ is a finite field extension of k. In particular,

$$\dim B/\mathfrak{m} = 0.$$

Using the given dimension formula for finitely generated k-algebras, we have

$$\mathrm{height}(\mathfrak{m}) + \dim B/\mathfrak{m} = \dim B.$$

Since $\dim B/\mathfrak{m} = 0$, it follows that

$$\mathrm{height}(\mathfrak{m}) = \dim B.$$

On the other hand,

$$\mathcal{O}_{X,p} \cong B_{\mathfrak{m}}.$$

The dimension of the local ring $B_{\mathfrak{m}}$ is the height of $\mathfrak{m}$:

$$\dim B_{\mathfrak{m}} = \mathrm{height}(\mathfrak{m}).$$

Hence

$$\dim \mathcal{O}_{X,p} = \mathrm{height}(\mathfrak{m}).$$

Therefore,

$$\dim \mathcal{O}_{X,p} = \dim B.$$

Since $U = \operatorname{Spec} B$ is a nonempty affine open subset of the variety X, we have

$$\dim B = \dim X.$$

Thus

$$\boxed{\dim \mathcal{O}_{X,p} = \dim X.}$$

2.2 Part (b): Codimension of a Closed Subset

Let $Y \subseteq X$ be a closed subset. We are given the definition

$$\operatorname{codim}(Y, X) = \inf_{Z \subseteq Y} \operatorname{codim}(Z, X),$$

where Z runs over the closed irreducible subsets of Y.

By Problem 1, every closed irreducible subset $Z \subseteq Y$ has a unique generic point $\eta_Z \in Z$ such that

$$\overline{\{\eta_Z\}} = Z.$$

Conversely, if $y \in Y$, then since Y is closed in X, we have

$$\overline{\{y\}} \subseteq Y.$$

The closure $\overline{\{y\}}$ is always irreducible, because it is the closure of a single point.

Thus the closed irreducible subsets of Y are exactly the closures

$$\overline{\{y\}}, \qquad y \in Y.$$

Now fix $y \in Y$. Choose an affine open neighbourhood

$$U = \operatorname{Spec} B \subseteq X$$

of y. Let y correspond to the prime ideal

$$\mathfrak{p} \subseteq B.$$

Then

$$\overline{\{y\}} \cap U = V(\mathfrak{p}).$$

The codimension of the irreducible closed subset $\overline{\{y\}}$ in X is locally the height of $\mathfrak{p}$:

$$\operatorname{codim}(\overline{\{y\}}, X) = \operatorname{height}(\mathfrak{p}).$$

Also,

$$\mathcal{O}_{X,y} \cong B_{\mathfrak{p}}.$$

Therefore

$$\dim \mathcal{O}_{X,y} = \dim B_{\mathfrak{p}} = \operatorname{height}(\mathfrak{p}).$$

Hence
$$\operatorname{codim}(\overline{\{y\}}, X) = \dim \mathcal{O}_{X,y}.$$

Taking the infimum over all $y \in Y$, we obtain

$$\operatorname{codim}(Y, X) = \inf_{y \in Y} \operatorname{codim}(\overline{\{y\}}, X) = \inf_{y \in Y} \dim \mathcal{O}_{X,y}.$$

Therefore
$$\boxed{\operatorname{codim}(Y, X) = \inf\{\dim \mathcal{O}_{X,y} \mid y \in Y\}.}$$

2.3 Part (c): Dimension Plus Codimension

Let $Y \subseteq X$ be a closed subset. We need to show that

$$\dim Y + \operatorname{codim}(Y, X) = \dim X.$$

Let $Z \subseteq Y$ be a closed irreducible subset. Let $\eta \in Z$ be its generic point, so that

$$Z = \overline{\{\eta\}}.$$

Choose an affine open subset
$$U = \operatorname{Spec} B \subseteq X$$
meeting Z. The point η corresponds to a prime ideal

$$\mathfrak{p} \subseteq B.$$

Then
$$Z \cap U = V(\mathfrak{p}).$$

The codimension of Z in X is

$$\operatorname{codim}(Z, X) = \operatorname{height}(\mathfrak{p}).$$

The dimension of Z is
$$\dim Z = \dim B/\mathfrak{p}.$$

Using the given formula,

$$\operatorname{height}(\mathfrak{p}) + \dim B/\mathfrak{p} = \dim B.$$

Therefore
$$\operatorname{codim}(Z, X) + \dim Z = \dim B.$$

Since $U = \operatorname{Spec} B$ is a nonempty affine open subset of X, we have

$$\dim B = \dim X.$$

Hence
$$\operatorname{codim}(Z, X) + \dim Z = \dim X.$$

Thus
$$\operatorname{codim}(Z,X) = \dim X - \dim Z.$$

Now take the infimum over all closed irreducible subsets $Z \subseteq Y$:
$$\operatorname{codim}(Y,X) = \inf_{Z\subseteq Y} \operatorname{codim}(Z,X).$$

Using the formula above,
$$\operatorname{codim}(Y,X) = \inf_{Z\subseteq Y} (\dim X - \dim Z).$$

Since $\dim X$ is fixed, this becomes
$$\operatorname{codim}(Y,X) = \dim X - \sup_{Z\subseteq Y} \dim Z.$$

But by definition,
$$\dim Y = \sup\{\dim Z \mid Z \subseteq Y \text{ closed irreducible}\}.$$

Therefore
$$\operatorname{codim}(Y,X) = \dim X - \dim Y.$$

Rearranging gives
$$\boxed{\dim Y + \operatorname{codim}(Y,X) = \dim X.}$$

2.4 Part (d): Nonempty Open Subsets Have the Same Dimension

Let $U \subseteq X$ be a nonempty open subset. We need to show that
$$\dim U = \dim X.$$

Since X is integral, every nonempty open subset of X is also integral. Therefore U is integral.

Since X is of finite type over k, the open subscheme U is also of finite type over k. Hence U is again a variety over k.

Because U is a nonempty open subset of the integral scheme X, U and X have the same function field:
$$K(U) = K(X).$$

Using Part (e), which states that the dimension of a variety is the transcendence degree of its function field, we get
$$\dim U = \operatorname{trdeg}_k K(U)$$

and
$$\dim X = \operatorname{trdeg}_k K(X).$$

Since

$$K(U) = K(X),$$

we conclude that

$$\dim U = \dim X.$$

Hence

$$\boxed{\dim U = \dim X.}$$

2.5 Part (e): Dimension Equals Transcendence Degree

We prove that

$$\dim X = \operatorname{trdeg}_k K(X).$$

Choose a nonempty affine open subset

$$U = \operatorname{Spec} B \subseteq X.$$

Since X is a variety over k, the ring B is a finitely generated k-algebra and a domain. The function field of X is the fraction field of B:

$$K(X) = \operatorname{Frac}(B).$$

By the commutative algebra fact given in the question,

$$\dim B = \operatorname{trdeg}_k \operatorname{Frac}(B).$$

Therefore

$$\dim B = \operatorname{trdeg}_k K(X).$$

Since $U = \operatorname{Spec} B$ is a nonempty affine open subset of the irreducible finite-type scheme X, it has the same dimension as X:

$$\dim U = \dim X.$$

But

$$\dim U = \dim B.$$

Thus

$$\dim X = \dim B = \operatorname{trdeg}_k K(X).$$

Therefore

$$\boxed{\dim X = \operatorname{trdeg}_k K(X).}$$

3 Problem 3: Counterexamples for General Schemes

Statement

For each property (a), (c), and (d) in Question 2, find a scheme X for which the property does not hold.

Solution

The statements in Problem 2 depend strongly on X being a variety, in particular on X being irreducible and equidimensional. For arbitrary schemes, the statements can fail.

We will use the same scheme to disprove all three properties.

Let

$$X = \operatorname{Spec} k \ \sqcup \ \mathbb{A}^1_k.$$

Equivalently,

$$X = \operatorname{Spec} k \ \sqcup \ \operatorname{Spec} k[t].$$

This is a scheme with two connected components:

$$\operatorname{Spec} k$$

and

$$\mathbb{A}^1_k = \operatorname{Spec} k[t].$$

Its dimension is

$$\dim X = 1,$$

because

$$\dim \operatorname{Spec} k = 0$$

and

$$\dim \mathbb{A}^1_k = 1.$$

The dimension of a disjoint union is the supremum of the dimensions of the components, so

$$\dim X = \max\{0, 1\} = 1.$$

3.1 Failure of Property (a)

Property (a) says that for any closed point $p \in X$,

$$\dim X = \dim \mathcal{O}_{X,p}.$$

Let p be the unique point of the component $\operatorname{Spec} k$. Then p is a closed point of X.

The local ring at p is

$$\mathcal{O}_{X,p} \cong k.$$

Therefore
$$\dim \mathcal{O}_{X,p} = 0.$$
But
$$\dim X = 1.$$
Hence
$$\dim \mathcal{O}_{X,p} = 0 \neq 1 = \dim X.$$
Therefore property (a) fails for this scheme:
$$\boxed{\dim X \neq \dim \mathcal{O}_{X,p}.}$$

3.2 Failure of Property (c)

Property (c) says that if $Y \subseteq X$ is closed, then
$$\dim Y + \operatorname{codim}(Y, X) = \dim X.$$
Let
$$Y = \operatorname{Spec} k \subseteq X$$
be the zero-dimensional connected component of X. Since connected components of a disjoint union are both open and closed, Y is closed in X.

We have
$$\dim Y = 0.$$
The only irreducible closed subset of Y is Y itself. Its generic point is the unique point $p \in \operatorname{Spec} k$.

The codimension of Y in X is
$$\operatorname{codim}(Y, X) = \dim \mathcal{O}_{X,p}.$$
But as above,
$$\mathcal{O}_{X,p} \cong k,$$
so
$$\dim \mathcal{O}_{X,p} = 0.$$
Therefore
$$\operatorname{codim}(Y, X) = 0.$$
Thus
$$\dim Y + \operatorname{codim}(Y, X) = 0 + 0 = 0.$$
But
$$\dim X = 1.$$
Hence
$$\dim Y + \operatorname{codim}(Y, X) = 0 \neq 1 = \dim X.$$
Therefore property (c) fails:
$$\boxed{\dim Y + \operatorname{codim}(Y, X) \neq \dim X.}$$

3.3 Failure of Property (d)

Property (d) says that if $U \subseteq X$ is a nonempty open subset, then

$$\dim U = \dim X.$$

Again take

$$U = \operatorname{Spec} k \subseteq X.$$

Since X is a disjoint union,

$$\operatorname{Spec} k$$

is both open and closed in X. Thus U is a nonempty open subset of X.

But

$$\dim U = 0,$$

whereas

$$\dim X = 1.$$

Therefore

$$\dim U = 0 \neq 1 = \dim X.$$

Hence property (d) fails:

$$\boxed{\dim U \neq \dim X.}$$

Conclusion

The scheme

$$X = \operatorname{Spec} k \sqcup \mathbb{A}^1_k$$

shows that properties (a), (c), and (d) from Problem 2 do not hold for arbitrary schemes.

The reason is that this scheme is not irreducible and not equidimensional. It has one component of dimension 0 and another component of dimension 1. Therefore closed points and open subsets lying entirely in the zero-dimensional component do not detect the full dimension of X.

4 Theory: Generic Points, Dimension, Codimension, and Varieties

This section collects the main theoretical background behind the three problems. The central ideas are generic points of irreducible closed subsets, the relation between prime ideals and closed subsets in affine schemes, the meaning of dimension and codimension, and the special dimension properties of varieties.

4.1 Generic Points and Irreducible Closed Subsets

A fundamental feature of schemes is that irreducible closed subsets usually have points which represent the whole subset.

Let X be a scheme and let $Z \subseteq X$ be a closed irreducible subset. A point

$$\eta \in Z$$

is called a **generic point** of Z if

$$\overline{\{\eta\}} = Z.$$

Thus η is not just an ordinary point of Z. It is a point whose closure is the whole subset Z. In this sense, η represents the entire irreducible closed subset.

In an affine scheme

$$X = \operatorname{Spec} A,$$

points are prime ideals

$$\mathfrak{p} \subseteq A.$$

The closure of the point $\mathfrak{p}$ is

$$\overline{\{\mathfrak{p}\}} = V(\mathfrak{p}),$$

where

$$V(\mathfrak{p}) = \{\mathfrak{q} \in \operatorname{Spec} A \mid \mathfrak{p} \subseteq \mathfrak{q}\}.$$

Therefore, in $\operatorname{Spec} A$, irreducible closed subsets are precisely the subsets of the form

$$V(\mathfrak{p}),$$

where $\mathfrak{p}$ is prime. The point $\mathfrak{p}$ is the generic point of $V(\mathfrak{p})$.

So we have the important dictionary

$$\boxed{\text{irreducible closed subsets of } \operatorname{Spec} A \longleftrightarrow \text{prime ideals of } A.}$$

The correspondence is

$$\mathfrak{p} \longmapsto V(\mathfrak{p}).$$

This is one of the main reasons schemes are useful: irreducible geometric pieces are encoded directly by prime ideals.

4.2 The T_0 Property and Uniqueness of Generic Points

Every scheme is a T_0-topological space. This means that if two points have the same closure, then they are equal.

Equivalently, for points $x, y \in X$,

$$\overline{\{x\}} = \overline{\{y\}} \quad \Longrightarrow \quad x = y.$$

Therefore, once a generic point exists, it is automatically unique.

Indeed, suppose that $\eta, \eta' \in Z$ are both generic points of a closed irreducible subset Z. Then

$$\overline{\{\eta\}} = Z$$

and

$$\overline{\{\eta'\}} = Z.$$

Hence

$$\overline{\{\eta\}} = \overline{\{\eta'\}}.$$

Since schemes are T_0, it follows that

$$\eta = \eta'.$$

Thus the generic point of an irreducible closed subset is unique.

The proof that every closed irreducible subset of a scheme has a generic point is local on affine open subsets. On an affine open

$$U = \operatorname{Spec} A,$$

the closed irreducible subsets are exactly the sets $V(\mathfrak{p})$, where $\mathfrak{p}$ is prime. Therefore the generic point is locally represented by that prime ideal.

4.3 Specialization and Generalization

Let X be a topological space. If $x, y \in X$, we say that y is a **specialization** of x if

$$y \in \overline{\{x\}}.$$

Equivalently, x is called a **generalization** of y.

In affine schemes this has a simple algebraic interpretation. Let

$$X = \operatorname{Spec} A.$$

Suppose $x = \mathfrak{p}$ and $y = \mathfrak{q}$, where $\mathfrak{p}, \mathfrak{q} \subseteq A$ are prime ideals. Then

$$\mathfrak{q} \in \overline{\{\mathfrak{p}\}}$$

if and only if

$$\mathfrak{p} \subseteq \mathfrak{q}.$$

Therefore,

$$\boxed{\mathfrak{p} \subseteq \mathfrak{q} \quad \Longleftrightarrow \quad \mathfrak{q} \text{ is a specialization of } \mathfrak{p}.}$$

The smaller prime ideal corresponds to the more generic point. The larger prime ideal corresponds to the more special point.

For example, in

$$\mathbb{A}^1_k = \operatorname{Spec} k[t],$$

the zero ideal (0) is the generic point of the whole affine line:

$$\overline{\{(0)\}} = \operatorname{Spec} k[t].$$

A maximal ideal of the form $(t-a)$, where $a \in k$, is a closed point. Its closure is just itself:

$$\overline{\{(t-a)\}} = \{(t-a)\}.$$

Thus the point (0) represents the whole affine line, while the maximal ideals represent ordinary closed points.

4.4 Dimension of Schemes

The dimension of a topological space X is defined as the supremum of the lengths of chains of irreducible closed subsets:

$$Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n.$$

The length of such a chain is n. Therefore,

$$\dim X = \sup\{n \mid \text{there exists a chain } Z_0 \subsetneq \cdots \subsetneq Z_n\}.$$

For affine schemes this agrees with Krull dimension. If

$$X = \operatorname{Spec} A,$$

then

$$\dim X = \dim A.$$

Here $\dim A$ is the supremum of the lengths of chains of prime ideals

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n.$$

Thus, in affine schemes, chains of irreducible closed subsets correspond to chains of prime ideals.

For example,

$$\dim \operatorname{Spec} k = 0,$$

because a field has exactly one prime ideal, namely (0).

Also,

$$\dim \mathbb{A}^1_k = \dim \operatorname{Spec} k[t] = 1.$$

A typical chain of prime ideals is

$$(0) \subsetneq (t-a).$$

Similarly,

$$\dim \mathbb{A}^n_k = \dim \operatorname{Spec} k[x_1, \ldots, x_n] = n.$$

The chain

$$(0) \subsetneq (x_1) \subsetneq (x_1, x_2) \subsetneq \cdots \subsetneq (x_1, \ldots, x_n)$$

has length n, showing that the dimension is at least n, and standard commutative algebra shows it is exactly n.

4.5 Varieties and Function Fields

In these problems, a variety over k is understood to be an integral scheme of finite type over k.

Thus a variety X has the following properties:

- X is irreducible.
- X is reduced.
- X is of finite type over k.

Because X is integral, it has a function field, denoted

$$K(X).$$

If

$$U = \operatorname{Spec} B \subseteq X$$

is any nonempty affine open subset, then B is a finitely generated k-algebra and a domain. Moreover,

$$K(X) = \operatorname{Frac}(B).$$

The dimension of a variety is connected to the transcendence degree of its function field by the formula

$$\boxed{\dim X = \operatorname{trdeg}_k K(X).}$$

This formula says that the dimension of X is the number of algebraically independent parameters needed to describe rational functions on X.

For example,

$$K(\mathbb{A}^n_k) = k(x_1, \dots, x_n),$$

and

$$\operatorname{trdeg}_k k(x_1, \dots, x_n) = n.$$

Therefore,

$$\dim \mathbb{A}^n_k = n.$$

This gives a bridge between geometry and field theory:

$$\boxed{\text{geometric dimension} = \text{number of independent transcendental parameters.}}$$

4.6 Local Rings and Local Dimension

Let X be a scheme and let $x \in X$. The local ring at x is denoted

$$\mathcal{O}_{X,x}.$$

This ring describes regular functions defined near x. In the affine case, if

$$X = \operatorname{Spec} A$$

and x corresponds to the prime ideal $\mathfrak{p} \subseteq A$, then

$$\mathcal{O}_{X,x} = A_{\mathfrak{p}}.$$

The dimension of this local ring is

$$\dim \mathcal{O}_{X,x} = \dim A_{\mathfrak{p}}.$$

For a prime ideal $\mathfrak{p}$, one has

$$\dim A_{\mathfrak{p}} = \operatorname{height}(\mathfrak{p}).$$

Therefore,

$$\boxed{\dim \mathcal{O}_{X,x} = \operatorname{height}(\mathfrak{p}).}$$

Geometrically, this number measures how far the point x is from the generic point. More precisely, it measures the maximum length of chains of specializations ending at x.

If X is a variety and $p \in X$ is a closed point, then locally p corresponds to a maximal ideal $\mathfrak{m}$. For finitely generated k-algebras, closed points have maximal possible height. Hence

$$\dim \mathcal{O}_{X,p} = \dim X.$$

So for closed points of a variety,

$$\boxed{\dim X = \dim \mathcal{O}_{X,p}.}$$

This is the statement of Problem 2(a).

The intuition is that a closed point is a most-special point. Starting from the generic point of the variety, one can specialize along a full chain down to a closed point. Therefore the local ring at a closed point detects the full dimension of the variety.

4.7 Codimension of Irreducible Closed Subsets

Let X be a variety and let

$$Z \subseteq X$$

be a closed irreducible subset. Let η_Z be the generic point of Z, so

$$\overline{\{\eta_Z\}} = Z.$$

The codimension of Z in X measures how many independent equations are needed, generically, to cut out Z inside X.

In affine language, suppose

$$X = \operatorname{Spec} B$$

and

$$Z = V(\mathfrak{p})$$

for a prime ideal $\mathfrak{p} \subseteq B$. Then

$$\eta_Z = \mathfrak{p}.$$

The codimension of Z in X is

$$\operatorname{codim}(Z, X) = \operatorname{height}(\mathfrak{p}).$$

But

$$\mathcal{O}_{X,\eta_Z} = B_{\mathfrak{p}}.$$

Therefore,

$$\dim \mathcal{O}_{X,\eta_Z} = \dim B_{\mathfrak{p}} = \operatorname{height}(\mathfrak{p}).$$

Hence

$$\boxed{\operatorname{codim}(Z, X) = \dim \mathcal{O}_{X,\eta_Z}.}$$

This is the fundamental relation between codimension and local dimension.

4.8 Codimension of Arbitrary Closed Subsets

If $Y \subseteq X$ is closed but not necessarily irreducible, then Y may have several irreducible components. Therefore one defines

$$\operatorname{codim}(Y, X) = \inf_{Z \subseteq Y} \operatorname{codim}(Z, X),$$

where Z runs over all closed irreducible subsets of Y.

By the existence of generic points, every closed irreducible subset $Z \subseteq Y$ is of the form

$$Z = \overline{\{y\}}$$

for a unique point $y \in Y$.

Conversely, if $y \in Y$, then since Y is closed,

$$\overline{\{y\}} \subseteq Y,$$

and $\overline{\{y\}}$ is irreducible.

Thus the closed irreducible subsets of Y correspond exactly to points of Y.

Therefore,

$$\operatorname{codim}(Y, X) = \inf_{y \in Y} \operatorname{codim}(\overline{\{y\}}, X).$$

Using the relation

$$\operatorname{codim}(\overline{\{y\}}, X) = \dim \mathcal{O}_{X,y},$$

we get

$$\boxed{\operatorname{codim}(Y, X) = \inf\{\dim \mathcal{O}_{X,y} \mid y \in Y\}.}$$

This is the statement of Problem 2(b).

The intuition is that the codimension of a reducible closed subset is the smallest codimension among its irreducible pieces. Equivalently, it is detected by the point of Y whose closure has the smallest codimension inside X.

4.9 The Dimension Formula for Varieties

The main commutative algebra fact used in the problem is the following.

Let B be a finitely generated k-algebra and a domain. If

$$\mathfrak{p} \subseteq B$$

is a prime ideal, then

$$\boxed{\operatorname{height}(\mathfrak{p}) + \dim B/\mathfrak{p} = \dim B.}$$

Geometrically, if

$$X = \operatorname{Spec} B$$

and

$$Z = V(\mathfrak{p}),$$

then

$$\operatorname{height}(\mathfrak{p}) = \operatorname{codim}(Z, X)$$

and

$$\dim B/\mathfrak{p} = \dim Z.$$

Thus the algebraic formula becomes the geometric formula

$$\boxed{\operatorname{codim}(Z, X) + \dim Z = \dim X.}$$

For a closed irreducible subset $Z \subseteq X$, this says

$$\dim Z + \operatorname{codim}(Z, X) = \dim X.$$

For an arbitrary closed subset $Y \subseteq X$, one takes the largest-dimensional irreducible component of Y. Since

$$\dim Y = \sup\{\dim Z \mid Z \subseteq Y \text{ closed irreducible}\},$$

and since

$$\operatorname{codim}(Y, X) = \inf\{\operatorname{codim}(Z, X) \mid Z \subseteq Y \text{ closed irreducible}\},$$

the same formula gives

$$\boxed{\dim Y + \operatorname{codim}(Y, X) = \dim X.}$$

This is Problem 2(c).

The intuition is that, inside a variety, dimension and codimension complement each other. If a closed subset has large dimension, it has small codimension. If it has small dimension, it has large codimension.

4.10 Nonempty Open Subsets of Varieties

Let X be an irreducible variety, and let

$$U \subseteq X$$

be a nonempty open subset.

Because X is irreducible, every nonempty open subset is dense. Therefore U contains the generic point of X. Hence U has the same function field as X:

$$K(U) = K(X).$$

Using the dimension-transcendence-degree formula, we have

$$\dim U = \operatorname{trdeg}_k K(U)$$

and

$$\dim X = \operatorname{trdeg}_k K(X).$$

Since

$$K(U) = K(X),$$

it follows that

$$\boxed{\dim U = \dim X.}$$

This is Problem 2(d).

The intuition is that removing a proper closed subset from an irreducible variety does not change the number of independent parameters. For example,

$$\mathbb{A}_k^2$$

has dimension 2, and the punctured affine plane

$$\mathbb{A}_k^2 \setminus \{0\}$$

also has dimension 2.

Open subsets may remove some special points, but they do not remove the generic point. Therefore they do not change the function field and hence do not change the dimension.

4.11 Dimension and Transcendence Degree

Let X be a variety over k. The equality

$$\dim X = \operatorname{trdeg}_k K(X)$$

is one of the most important dimension formulas in algebraic geometry.

Choose a nonempty affine open subset

$$U = \operatorname{Spec} B \subseteq X.$$

Since X is a variety, B is a finitely generated k-algebra and a domain.

The function field of X is

$$K(X) = \operatorname{Frac}(B).$$

By the commutative algebra fact,

$$\dim B = \operatorname{trdeg}_k \operatorname{Frac}(B).$$

Hence

$$\dim B = \operatorname{trdeg}_k K(X).$$

Since U is a nonempty open subset of the irreducible variety X, it has the same dimension as X:

$$\dim U = \dim X.$$

But

$$\dim U = \dim B.$$

Therefore,

$$\boxed{\dim X = \operatorname{trdeg}_k K(X).}$$

This is Problem 2(e).

This formula explains why algebraic curves have dimension 1, algebraic surfaces have dimension 2, and so on. For instance,

$$K(\mathbb{A}^1_k) = k(t),$$

so

$$\operatorname{trdeg}_k k(t) = 1.$$

Therefore

$$\dim \mathbb{A}^1_k = 1.$$

Similarly,

$$K(\mathbb{A}^2_k) = k(x, y),$$

and

$$\operatorname{trdeg}_k k(x, y) = 2.$$

Therefore

$$\dim \mathbb{A}^2_k = 2.$$

4.12 Why the Variety Assumption Matters

The statements in Problem 2 rely heavily on X being a variety. In particular, they use the fact that X is irreducible and finite type over k.

If X is an arbitrary scheme, then the statements can fail. The simplest reason is that X may have several connected components or irreducible components of different dimensions.

For example, consider

$$X = \operatorname{Spec} k \sqcup \mathbb{A}^1_k.$$

Equivalently,

$$X = \operatorname{Spec} k \sqcup \operatorname{Spec} k[t].$$

Then X has two connected components. One component is

$$\operatorname{Spec} k,$$

which has dimension 0, and the other component is

$$\mathbb{A}^1_k,$$

which has dimension 1.

Therefore,

$$\dim X = 1.$$

Now let p be the unique point of the component $\operatorname{Spec} k$. Then p is a closed point of X, and

$$\mathcal{O}_{X,p} \cong k.$$

Hence

$$\dim \mathcal{O}_{X,p} = 0.$$

But

$$\dim X = 1.$$

Thus

$$\dim \mathcal{O}_{X,p} \neq \dim X.$$

So property 2(a) fails.

Now take

$$Y = \operatorname{Spec} k \subseteq X.$$

This subset is closed, because it is a connected component of a disjoint union. We have

$$\dim Y = 0.$$

The codimension of Y in X is also 0, because Y is itself an irreducible component of X. Hence

$$\dim Y + \operatorname{codim}(Y, X) = 0 + 0 = 0.$$

But

$$\dim X = 1.$$

Therefore

$$\dim Y + \operatorname{codim}(Y, X) \neq \dim X.$$

So property 2(c) fails.

Finally, take

$$U = \operatorname{Spec} k \subseteq X.$$

Since it is a connected component of a disjoint union, it is both open and closed. Thus U is a nonempty open subset of X. However,

$$\dim U = 0,$$

whereas

$$\dim X = 1.$$

Therefore

$$\dim U \neq \dim X.$$

So property 2(d) fails.

The common reason for these failures is that X is not irreducible and not equidimensional. It has a zero-dimensional component and a one-dimensional component. Points and open subsets lying entirely inside the zero-dimensional component cannot detect the one-dimensional part of X.

4.13 Summary Table

Concept	**Algebraic Form**	**Geometric Meaning**
Point of $\operatorname{Spec} A$	Prime ideal $\mathfrak{p} \subseteq A$	An irreducible algebraic condition
Closure of a point	$\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$	The irreducible closed subset represented by $\mathfrak{p}$
Generic point	$\mathfrak{p}$ for $V(\mathfrak{p})$	A point whose closure is an irreducible closed subset
Closed point	Maximal ideal $\mathfrak{m} \subseteq A$	An ordinary geometric point
Dimension of $\operatorname{Spec} A$	Krull dimension $\dim A$	Maximum length of chains of irreducible closed subsets
Local dimension	$\dim \mathcal{O}_{X,x} = \operatorname{height}(\mathfrak{p})$	Codimension of the closure of x
Codimension of $V(\mathfrak{p})$	$\operatorname{codim}(V(\mathfrak{p}), X) = \operatorname{height}(\mathfrak{p})$	Number of equations needed generically
Dimension of a variety	$\dim X = \operatorname{trdeg}_k K(X)$	Number of independent parameters
Closed subset dimension formula	$\dim Y + \operatorname{codim}(Y, X) = \dim X$	Dimension and codimension complement each other
Nonempty open subset of a variety	$\dim U = \dim X$	Removing a proper closed subset does not change dimension

4.14 Main Ideas to Remember

The generic point of an irreducible closed subset Z is the unique point η such that

$$\overline{\{\eta\}} = Z.$$

In an affine scheme,

$$\overline{\{\mathfrak{p}\}} = V(\mathfrak{p}).$$

Thus prime ideals are exactly the generic points of irreducible closed subsets.

For a variety X,

$$\dim X = \operatorname{trdeg}_k K(X).$$

For a point $y \in X$,

$$\dim \mathcal{O}_{X,y} = \operatorname{codim}(\overline{\{y\}}, X).$$

For a closed subset $Y \subseteq X$,

$$\operatorname{codim}(Y, X) = \inf\{\dim \mathcal{O}_{X,y} \mid y \in Y\}.$$

For a closed subset $Y \subseteq X$ of a variety,

$$\dim Y + \operatorname{codim}(Y, X) = \dim X.$$

For a nonempty open subset $U \subseteq X$ of a variety,

$$\dim U = \dim X.$$

These statements can fail for arbitrary schemes because arbitrary schemes may be reducible, disconnected, or have components of different dimensions.

5 Examples: Generic Points, Dimension, Codimension, and Varieties

This section gives concrete examples illustrating generic points, local dimensions, codimensions, the dimension formula for varieties, nonempty open subsets, transcendence degree, and counterexamples for arbitrary schemes.

5.1 Generic Point of the Affine Line

Let

$$X = \mathbb{A}^1_k = \operatorname{Spec} k[t].$$

The prime ideals of $k[t]$ are the zero ideal

$$(0)$$

and maximal ideals such as

$$(t-a), \qquad a \in k.$$

The closure of the point (0) is

$$\overline{\{(0)\}} = V(0) = \operatorname{Spec} k[t] = \mathbb{A}^1_k.$$

Therefore (0) is the generic point of the whole affine line.

On the other hand,

$$\overline{\{(t-a)\}} = V(t-a) = \{(t-a)\}.$$

Thus $(t-a)$ is a closed point.

Hence

$$\boxed{(0) \text{ is the generic point of } \mathbb{A}^1_k}$$

and

$$\boxed{(t-a) \text{ is a closed point of } \mathbb{A}^1_k.}$$

5.2 Generic Point of a Closed Irreducible Subset in $\mathbb{A}^2_k$

Let

$$X = \mathbb{A}^2_k = \operatorname{Spec} k[x,y].$$

Consider the closed subset

$$Z = V(y - x^2).$$

Geometrically, Z is the parabola

$$y = x^2.$$

The ideal

$$(y - x^2)$$

is prime, because

$$k[x,y]/(y-x^2) \cong k[x],$$

and $k[x]$ is a domain.

Therefore Z is irreducible. Its generic point is the prime ideal

$$\eta_Z = (y - x^2).$$

Indeed,

$$\overline{\{(y-x^2)\}} = V(y-x^2) = Z.$$

Thus

$$\boxed{(y-x^2) \text{ is the generic point of the parabola } V(y-x^2).}$$

5.3 Closed Point and Local Dimension in $\mathbb{A}^2_k$

Let

$$X = \mathbb{A}^2_k = \operatorname{Spec} k[x,y].$$

Assume that k is algebraically closed. A closed point of X is given by a maximal ideal

$$\mathfrak{m} = (x-a, y-b),$$

corresponding to the ordinary point

$$(a, b) \in \mathbb{A}^2_k.$$

The local ring at this point is

$$\mathcal{O}_{X,\mathfrak{m}} = k[x, y]_{(x-a,y-b)}.$$

Its dimension is the height of the maximal ideal:

$$\dim k[x, y]_{(x-a,y-b)} = \text{height}(x - a, y - b).$$

Since

$$\text{height}(x - a, y - b) = 2,$$

we get

$$\dim \mathcal{O}_{X,\mathfrak{m}} = 2.$$

Also,

$$\dim X = \dim k[x, y] = 2.$$

Therefore

$$\boxed{\dim \mathcal{O}_{X,\mathfrak{m}} = \dim X = 2.}$$

This illustrates the fact that for a variety X, every closed point $p \in X$ satisfies

$$\dim \mathcal{O}_{X,p} = \dim X.$$

5.4 Codimension of a Line in $\mathbb{A}^2_k$

Let

$$X = \mathbb{A}^2_k = \operatorname{Spec} k[x, y].$$

Let

$$Y = V(y).$$

Geometrically, Y is the x-axis.

The ideal (y) is prime because

$$k[x, y]/(y) \cong k[x],$$

and $k[x]$ is a domain.

Therefore Y is irreducible, and its generic point is

$$\eta_Y = (y).$$

Now

$$\text{height}(y) = 1.$$

Hence

$$\operatorname{codim}(Y, X) = 1.$$

Also,

$$\dim Y = \dim k[x] = 1,$$

and

$$\dim X = 2.$$

Therefore

$$\dim Y + \operatorname{codim}(Y, X) = 1 + 1 = 2 = \dim X.$$

Thus

$$\boxed{\dim Y + \operatorname{codim}(Y, X) = \dim X.}$$

5.5 Codimension of the Origin in $\mathbb{A}_k^2$

Let

$$X = \mathbb{A}_k^2 = \operatorname{Spec} k[x, y].$$

Let

$$Y = V(x, y).$$

This is the origin $(0, 0)$.

The ideal (x, y) is maximal, so Y is a closed point.

We have

$$\dim Y = 0.$$

Also,

$$\operatorname{codim}(Y, X) = \operatorname{height}(x, y) = 2.$$

Since

$$\dim X = 2,$$

we obtain

$$\dim Y + \operatorname{codim}(Y, X) = 0 + 2 = 2 = \dim X.$$

Thus

$$\boxed{\dim V(x, y) + \operatorname{codim}(V(x, y), \mathbb{A}_k^2) = 2.}$$

5.6 Reducible Closed Subset in $\mathbb{A}_k^2$

Let

$$X = \mathbb{A}_k^2 = \operatorname{Spec} k[x, y].$$

Consider

$$Y = V(xy).$$

Since

$$xy = 0$$

means

$$x = 0 \quad \text{or} \quad y = 0,$$

we have

$$Y = V(x) \cup V(y).$$

Thus Y is the union of the y-axis and the x-axis.

The irreducible components are

$$V(x)$$

and

$$V(y).$$

Each component has dimension 1, so

$$\dim Y = 1.$$

Each component has codimension 1 in $\mathbb{A}^2_k$, because

$$\text{height}(x) = 1, \qquad \text{height}(y) = 1.$$

Therefore

$$\operatorname{codim}(Y, X) = 1.$$

Hence

$$\dim Y + \operatorname{codim}(Y, X) = 1 + 1 = 2 = \dim X.$$

So even though Y is reducible, the formula still works for closed subsets of a variety:

$$\boxed{\dim Y + \operatorname{codim}(Y, X) = \dim X.}$$

5.7 Closed Subset Contained in a Larger Irreducible Component

Let

$$X = \mathbb{A}^2_k = \operatorname{Spec} k[x, y].$$

Let

$$Y = V(x) \cup V(x, y).$$

Here $V(x)$ is the y-axis, while $V(x, y)$ is the origin.

But

$$V(x, y) \subseteq V(x),$$

so actually

$$Y = V(x).$$

Therefore

$$\dim Y = 1$$

and

$$\operatorname{codim}(Y, X) = 1.$$

This example shows that a smaller-dimensional closed subset contained inside a larger irreducible component does not change the dimension of the whole closed subset.

The dimension is controlled by the largest irreducible components.

5.8 Closed Subset with Components of Different Dimensions

Let
$$X = \mathbb{A}^3_k = \operatorname{Spec} k[x, y, z].$$
Let
$$Y = V(x) \cup V(y, z).$$
Here $V(x)$ is a plane, so
$$\dim V(x) = 2.$$
Also, $V(y, z)$ is the x-axis, so
$$\dim V(y, z) = 1.$$
Therefore
$$\dim Y = 2.$$
Now
$$\operatorname{codim}(V(x), X) = 1,$$
while
$$\operatorname{codim}(V(y, z), X) = 2.$$
The codimension of Y is the infimum of the codimensions of its irreducible pieces:
$$\operatorname{codim}(Y, X) = \inf\{1, 2\} = 1.$$
Thus
$$\dim Y + \operatorname{codim}(Y, X) = 2 + 1 = 3 = \dim X.$$
Hence
$$\boxed{\dim Y = 2, \qquad \operatorname{codim}(Y, X) = 1.}$$
The largest-dimensional component determines $\dim Y$, and the smallest-codimension component determines $\operatorname{codim}(Y, X)$.

5.9 Nonempty Open Subset of $\mathbb{A}^2_k$

Let
$$X = \mathbb{A}^2_k.$$
Then
$$\dim X = 2.$$

Remove the origin:

$$U = \mathbb{A}^2_k \setminus \{(0,0)\}.$$

This is a nonempty open subset of X.

Because X is irreducible and U is nonempty open, we have

$$\dim U = \dim X = 2.$$

Therefore

$$\boxed{\dim(\mathbb{A}^2_k \setminus \{(0,0)\}) = 2.}$$

Although one point was removed, the dimension did not change.

5.10 Distinguished Open Subset

Let

$$X = \operatorname{Spec} k[x,y].$$

Consider the distinguished open subset

$$D(x) = X \setminus V(x).$$

Geometrically, this is the part of $\mathbb{A}^2_k$ where

$$x \neq 0.$$

Algebraically,

$$D(x) = \operatorname{Spec} k[x,y]_x.$$

Since $D(x)$ is a nonempty open subset of the irreducible variety X, we have

$$\dim D(x) = \dim X = 2.$$

Also,

$$K(D(x)) = k(x,y) = K(X).$$

Thus

$$\boxed{\dim D(x) = 2.}$$

5.11 Dimension Equals Transcendence Degree

Let

$$X = \mathbb{A}^3_k = \operatorname{Spec} k[x,y,z].$$

The function field is

$$K(X) = k(x,y,z).$$

The elements x, y, z are algebraically independent over k, so

$$\operatorname{trdeg}_k k(x, y, z) = 3.$$

Therefore

$$\dim X = 3.$$

Thus

$$\boxed{\dim \mathbb{A}^3_k = \operatorname{trdeg}_k k(x, y, z) = 3.}$$

5.12 Dimension of a Hypersurface

Let

$$X = V(z - x^2 - y^2) \subseteq \mathbb{A}^3_k.$$

Then

$$X = \operatorname{Spec} k[x, y, z]/(z - x^2 - y^2).$$

But

$$k[x, y, z]/(z - x^2 - y^2) \cong k[x, y].$$

Therefore

$$X \cong \mathbb{A}^2_k.$$

Hence

$$\dim X = 2.$$

The function field is

$$K(X) = k(x, y).$$

Therefore

$$\operatorname{trdeg}_k K(X) = 2.$$

Hence

$$\boxed{\dim X = \operatorname{trdeg}_k K(X) = 2.}$$

5.13 Codimension of a Hypersurface in $\mathbb{A}^3_k$

Let

$$X = \mathbb{A}^3_k = \operatorname{Spec} k[x, y, z].$$

Let

$$Y = V(z - x^2 - y^2).$$

As above,

$$Y \cong \mathbb{A}^2_k,$$

so

$$\dim Y = 2.$$

Since Y is cut out by one nonzero prime equation, we have

$$\operatorname{codim}(Y, X) = 1.$$

Then

$$\dim Y + \operatorname{codim}(Y, X) = 2 + 1 = 3 = \dim X.$$

Thus

$$\boxed{\dim Y + \operatorname{codim}(Y, X) = \dim X.}$$

5.14 Counterexample for Arbitrary Schemes: Failure of Local Dimension Formula

Let

$$X = \operatorname{Spec} k \sqcup \mathbb{A}^1_k.$$

Equivalently,

$$X = \operatorname{Spec} k \sqcup \operatorname{Spec} k[t].$$

Then X has one component of dimension 0 and one component of dimension 1. Therefore

$$\dim X = 1.$$

Let p be the unique point of the component $\operatorname{Spec} k$. Then

$$\mathcal{O}_{X,p} = k.$$

Hence

$$\dim \mathcal{O}_{X,p} = 0.$$

But

$$\dim X = 1.$$

Therefore

$$\boxed{\dim \mathcal{O}_{X,p} \neq \dim X.}$$

So the statement

$$\dim \mathcal{O}_{X,p} = \dim X$$

for closed points p fails for arbitrary schemes.

5.15 Counterexample for Arbitrary Schemes: Open Subset with Smaller Dimension

Again let

$$X = \operatorname{Spec} k \sqcup \mathbb{A}^1_k.$$

The subset

$$U = \operatorname{Spec} k$$

is open, because it is a connected component of a disjoint union.

But

$$\dim U = 0,$$

whereas

$$\dim X = 1.$$

Therefore

$$\boxed{\dim U \neq \dim X.}$$

So the statement that every nonempty open subset has the same dimension as X fails for arbitrary schemes.

5.16 Counterexample for Arbitrary Schemes: Failure of the Dimension-Codimension Formula

Let

$$X = \operatorname{Spec} k \sqcup \mathbb{A}^1_k.$$

Let

$$Y = \operatorname{Spec} k \subseteq X.$$

The subset Y is closed in X, because it is a connected component.

We have

$$\dim Y = 0.$$

Also,

$$\operatorname{codim}(Y, X) = 0,$$

because Y is itself an irreducible component of X.

Therefore

$$\dim Y + \operatorname{codim}(Y, X) = 0 + 0 = 0.$$

But

$$\dim X = 1.$$

Hence

$$\boxed{\dim Y + \operatorname{codim}(Y, X) \neq \dim X.}$$

So the dimension-codimension formula fails for arbitrary schemes.

5.17 Summary of the Examples

Example	Object	Main Point
1	$\mathbb{A}^1_k$	(0) is the generic point
2	$V(y-x^2) \subseteq \mathbb{A}^2_k$	A prime ideal gives a generic point
3	Closed point in $\mathbb{A}^2_k$	Local dimension equals global dimension
4	$V(y) \subseteq \mathbb{A}^2_k$	A line has codimension 1
5	$V(x,y) \subseteq \mathbb{A}^2_k$	The origin has codimension 2
6	$V(xy) \subseteq \mathbb{A}^2_k$	Reducible closed subset
8	$V(x) \cup V(y,z) \subseteq \mathbb{A}^3_k$	Largest component controls dimension
9	$\mathbb{A}^2_k \setminus \{(0,0)\}$	Nonempty open subset has same dimension
11	$\mathbb{A}^3_k$	Dimension equals transcendence degree
$14--16$	$\operatorname{Spec} k \sqcup \mathbb{A}^1_k$	Failures for arbitrary schemes

6 Diagrams, Charts, and Tables: Generic Points, Dimension, and Codimension

This section collects visual summaries for the main ideas: generic points, specialization, local dimension, codimension, open subsets, transcendence degree, and counterexamples for arbitrary schemes.

6.1 Generic Point Diagram

For an affine scheme

$$X = \operatorname{Spec} A,$$

a point is a prime ideal

$$\mathfrak{p} \subseteq A.$$

Its closure is

$$\overline{\{\mathfrak{p}\}} = V(\mathfrak{p}).$$

Thus the generic-point relation can be summarized as follows:

$$\begin{gathered} \text{prime ideal } \mathfrak{p} \\ \Downarrow \text{ point of } \operatorname{Spec} A \\ \mathfrak{p} \in \operatorname{Spec} A \\ \Downarrow \text{ take closure} \\ \overline{\{\mathfrak{p}\}} = V(\mathfrak{p}) \\ \Downarrow \\ \text{closed irreducible subset} \end{gathered}$$

Therefore

$$\boxed{\mathfrak{p} \text{ is the generic point of } V(\mathfrak{p}).}$$

6.2 Specialization Diagram in $\mathbb{A}^1_k$

Let

$$\mathbb{A}^1_k = \operatorname{Spec} k[t].$$

The generic point is

$$(0).$$

Closed points are maximal ideals such as

$$(t-a), \qquad a \in k,$$

when k is algebraically closed.

The specialization diagram has the form

$$\begin{array}{ccccc} & (0) & & & \text{generic point of } \mathbb{A}^1_k \\ \swarrow & \downarrow & \searrow & & \\ (t-a_1) & (t-a_2) & (t-a_3) & \cdots & \text{closed points} \end{array}$$

The arrows mean specialization. Algebraically,

$$(0) \subseteq (t-a).$$

Thus the smaller prime ideal is more generic, and the larger prime ideal is more special.

6.3 Specialization Chain in $\mathbb{A}^2_k$

Let

$$X = \mathbb{A}^2_k = \operatorname{Spec} k[x,y].$$

A typical chain of prime ideals is

$$(0) \subsetneq (y) \subsetneq (x-a, y).$$

This gives the specialization chain

$$\begin{array}{c} (0) \\ \downarrow \\ (y) \\ \downarrow \\ (x-a,y) \end{array}$$

The geometric meaning is:

Prime ideal	**Geometric subset**	**Dimension**
(0)	$\mathbb{A}^2_k$	2
(y)	x-axis	1
$(x-a,y)$	point $(a,0)$	0

So a maximal chain has length 2, and hence

$$\dim \mathbb{A}^2_k = 2.$$

6.4 Dimension and Codimension Table in $\mathbb{A}_k^2$

Let
$$X = \mathbb{A}_k^2.$$

Then $\dim X = 2$. The following table shows how dimension and codimension balance each other.

Closed subset Y	**Ideal**	$\dim Y$	$\operatorname{codim}(Y, X)$
X	(0)	2	0
line $V(y)$	(y)	1	1
origin $V(x, y)$	(x, y)	0	2

In every row,
$$\dim Y + \operatorname{codim}(Y, X) = 2 = \dim X.$$

Thus the table illustrates
$$\boxed{\dim Y + \operatorname{codim}(Y, X) = \dim X.}$$

6.5 Dimension–Codimension Balance Chart in $\mathbb{A}_k^3$

Let
$$X = \mathbb{A}_k^3.$$

Then
$$\dim X = 3.$$

Typical closed subsets have the following dimensions and codimensions:

Object	**Example**	dim	codim
whole space	$\mathbb{A}_k^3$	3	0
surface	$V(z - x^2 - y^2)$	2	1
curve	$V(y, z)$	1	2
point	$V(x, y, z)$	0	3

The general pattern is:

$\dim Y$	3	2	1	0
$\operatorname{codim}(Y, X)$	0	1	2	3

Thus for $X = \mathbb{A}_k^3$,
$$\dim Y + \operatorname{codim}(Y, X) = 3.$$

6.6 Reducible Closed Subset Diagram

Let
$$Y = V(xy) \subseteq \mathbb{A}_k^2.$$

Since

$$xy = 0$$

means

$$x = 0 \qquad \text{or} \qquad y = 0,$$

we have

$$Y = V(x) \cup V(y).$$

Diagrammatically:

$$\begin{array}{ccc} & V(xy) & \\ \swarrow & & \searrow \\ V(x) & & V(y) \\ y\text{-axis} & & x\text{-axis} \end{array}$$

The generic points of the irreducible components are

$$(x) \qquad \text{and} \qquad (y).$$

Thus:

Component	**Generic point**	**Dimension**
$V(x)$	(x)	1
$V(y)$	(y)	1

Therefore

$$\dim V(xy) = 1.$$

6.7 Closed Subset with Components of Different Dimensions

Let

$$X = \mathbb{A}^3_k$$

and

$$Y = V(x) \cup V(y, z).$$

Here $V(x)$ is a plane, while $V(y, z)$ is a line.

$$\begin{array}{ccc} & Y = V(x) \cup V(y,z) & \\ \swarrow & & \searrow \\ V(x) & & V(y,z) \\ \text{plane} & & \text{line} \\ \dim = 2 & & \dim = 1 \\ \text{codim} = 1 & & \text{codim} = 2 \end{array}$$

Therefore,

$$\dim Y = 2,$$

because the largest-dimensional component is the plane.

Also,

$$\operatorname{codim}(Y, X) = 1,$$

because the smallest codimension is the codimension of the plane.

Hence

$$\dim Y + \operatorname{codim}(Y, X) = 2 + 1 = 3 = \dim X.$$

6.8 Local Ring and Codimension Table

Let

$$X = \operatorname{Spec} A.$$

Let $x \in X$ correspond to a prime ideal

$$\mathfrak{p} \subseteq A.$$

Then

$$\mathcal{O}_{X,x} = A_{\mathfrak{p}}.$$

Also,

$$\dim \mathcal{O}_{X,x} = \operatorname{height}(\mathfrak{p}).$$

The following table summarizes the relation:

Point x	**Prime** $\mathfrak{p}$	$\overline{\{x\}}$	$\dim \mathcal{O}_{X,x}$
generic point	(0)	X	0
generic point of divisor	(f)	$V(f)$	1
closed point in $\mathbb{A}^2_k$	$(x-a, y-b)$	$\{(a,b)\}$	2

Thus:

$$\boxed{\dim \mathcal{O}_{X,x} = \operatorname{codim}(\overline{\{x\}}, X).}$$

6.9 Open Subset Diagram

Let

$$X = \mathbb{A}^2_k$$

and

$$U = \mathbb{A}^2_k \setminus \{(0,0)\}.$$

Then:

$$X = \mathbb{A}^2_k$$

$$\Downarrow \text{ remove one closed point}$$

$$U = \mathbb{A}^2_k \setminus \{(0,0)\}$$

Although the origin has been removed, the function field remains the same:

$$K(U) = K(X) = k(x, y).$$

Therefore

$$\dim U = \dim X = 2.$$

This can be summarized as:

Space	**Function field**	**Dimension**
$\mathbb{A}^2_k$	$k(x, y)$	2
$\mathbb{A}^2_k \setminus \{(0, 0)\}$	$k(x, y)$	2

So nonempty open subsets of varieties preserve dimension.

6.10 Transcendence Degree Chart

For affine spaces:

Variety	**Function field**	**Dimension**
$\mathbb{A}^1_k$	$k(t)$	1
$\mathbb{A}^2_k$	$k(x, y)$	2
$\mathbb{A}^3_k$	$k(x, y, z)$	3
$\mathbb{A}^n_k$	$k(x_1, \ldots, x_n)$	n

In general, for a variety X,

$$\boxed{\dim X = \operatorname{trdeg}_k K(X).}$$

This means that dimension counts the number of algebraically independent parameters in the function field.

6.11 Counterexample Diagram: Disconnected Scheme

Let

$$X = \operatorname{Spec} k \sqcup \mathbb{A}^1_k.$$

Then:

$$X \quad = \quad \underbrace{\operatorname{Spec} k}_{\dim 0} \sqcup \underbrace{\mathbb{A}^1_k}_{\dim 1}$$

Therefore

$$\dim X = 1.$$

However, the subset

$$U = \operatorname{Spec} k$$

is open and closed in X, because it is a connected component.

But

$$\dim U = 0.$$

Hence

$$\boxed{\dim U \neq \dim X.}$$

This shows why the variety assumption matters.

6.12 Failure Table for Arbitrary Schemes

For

$$X = \operatorname{Spec} k \sqcup \mathbb{A}^1_k,$$

the statements from varieties fail as follows:

Property	Expected for varieties	Here	Result
closed point local dimension	$\dim \mathcal{O}_{X,p} = \dim X$	$0 \neq 1$	fails
closed subset formula	$\dim Y + \operatorname{codim}(Y, X) = \dim X$	$0 + 0 \neq 1$	fails
open subset dimension	$\dim U = \dim X$	$0 \neq 1$	fails

The reason is that X has components of different dimensions.

6.13 Master Summary Table

Topic	Formula or Diagram	Meaning
generic point	$\overline{\{\eta\}} = Z$	η represents Z
affine closure	$\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$	prime ideals give irreducible subsets
local dimension	$\dim \mathcal{O}_{X,x} = \operatorname{height}(\mathfrak{p})$	codimension of $\overline{\{x\}}$
dimension formula	$\dim Y + \operatorname{codim}(Y, X) = \dim X$	dimension and codimension complement
function field	$\dim X = \operatorname{trdeg}_k K(X)$	dimension equals number of parameters
open subset	$\dim U = \dim X$	nonempty opens preserve dimension
counterexample	$\operatorname{Spec} k \sqcup \mathbb{A}^1_k$	non-equidimensional schemes fail

7 Problem 1 on Concrete Spaces: Generic Points

Problem 1 says that if X is a scheme and $Z \subseteq X$ is a closed irreducible subset, then there exists a unique point

$$\eta \in Z$$

such that

$$\overline{\{\eta\}} = Z.$$

This point is called the **generic point** of Z.

The most important rule is the following:

$$\boxed{\text{In } \operatorname{Spec} A, \text{ the generic point of } V(\mathfrak{p}) \text{ is } \mathfrak{p}.}$$

Here $\mathfrak{p} \subseteq A$ is a prime ideal.

7.1 The Affine Line $\mathbb{A}^1_k$

Let

$$X = \mathbb{A}^1_k = \operatorname{Spec} k[t].$$

The points of X are the prime ideals of $k[t]$. If k is algebraically closed, then the points are

$$(0)$$

and

$$(t-a), \qquad a \in k.$$

The point (0) is not closed. Its closure is

$$\overline{\{(0)\}} = V(0) = \operatorname{Spec} k[t] = \mathbb{A}^1_k.$$

Therefore

$$\boxed{(0) \text{ is the generic point of } \mathbb{A}^1_k.}$$

Thus, in this example,

$$Z = X = \mathbb{A}^1_k$$

is closed and irreducible, and its generic point is

$$\eta = (0).$$

The specialization picture is

$$\begin{array}{ccccccc} & (0) & & & & \text{generic point of } \mathbb{A}^1_k \\ \swarrow & \downarrow & \searrow & & & \\ (t-a_1) & (t-a_2) & (t-a_3) & \cdots & & \text{closed points} \end{array}$$

The inclusion

$$(0) \subseteq (t-a)$$

means that the generic point (0) specializes to the closed point $(t-a)$.

7.2 A Closed Point in $\mathbb{A}^1_k$

Again let

$$X = \mathbb{A}^1_k = \operatorname{Spec} k[t].$$

Take

$$Z = \{(t-a)\} = V(t-a).$$

This is a closed irreducible subset consisting of one point.

Its generic point is

$$\eta = (t-a).$$

Indeed,

$$\overline{\{(t-a)\}} = \{(t-a)\} = V(t-a).$$

Therefore

$$\boxed{(t-a) \text{ is the generic point of the closed point } V(t-a).}$$

So a closed point is its own generic point.

7.3 The Affine Plane $\mathbb{A}^2_k$

Let

$$X = \mathbb{A}^2_k = \operatorname{Spec} k[x,y].$$

Since $k[x,y]$ is a domain, the affine plane is irreducible. The generic point of the whole plane is the zero ideal:

$$\eta = (0).$$

Indeed,

$$\overline{\{(0)\}} = V(0) = \operatorname{Spec} k[x,y] = \mathbb{A}^2_k.$$

Hence

$$\boxed{(0) \text{ is the generic point of } \mathbb{A}^2_k.}$$

A typical chain of specializations is

$$(0) \subsetneq (y) \subsetneq (x-a, y).$$

Geometrically, this means

Prime ideal	**Geometric meaning**
(0)	generic point of the whole plane
(y)	generic point of the x-axis
$(x-a, y)$	closed point $(a, 0)$

So the specialization diagram is

$$\begin{array}{c} (0) \\ \downarrow \\ (y) \\ \downarrow \\ (x-a, y) \end{array}$$

7.4 A Line in $\mathbb{A}^2_k$

Let

$$X = \mathbb{A}^2_k = \operatorname{Spec} k[x,y].$$

Consider the closed subset
$$Z = V(y).$$

Geometrically, Z is the x-axis.

The ideal (y) is prime because
$$k[x, y]/(y) \cong k[x],$$
and $k[x]$ is a domain. Therefore $Z = V(y)$ is closed and irreducible.

Its generic point is
$$\eta = (y).$$

Indeed,
$$\overline{\{(y)\}} = V(y) = Z.$$

Hence
$$\boxed{(y) \text{ is the generic point of the } x\text{-axis.}}$$

The closed points of this line are ideals of the form
$$(x - a, y).$$

Since
$$(y) \subsetneq (x - a, y),$$
the closed point $(x - a, y)$ is a specialization of the generic point (y).

7.5 A Parabola in $\mathbb{A}^2_k$

Let
$$X = \mathbb{A}^2_k = \operatorname{Spec} k[x, y].$$

Consider
$$Z = V(y - x^2).$$

Geometrically, this is the parabola
$$y = x^2.$$

The ideal
$$(y - x^2)$$
is prime because
$$k[x, y]/(y - x^2) \cong k[x],$$
which is a domain.

Therefore Z is closed and irreducible.

Its generic point is
$$\eta = (y - x^2).$$

Indeed,
$$\overline{\{(y - x^2)\}} = V(y - x^2) = Z.$$

Thus

$$\boxed{(y - x^2) \text{ is the generic point of the parabola } y = x^2.}$$

A closed point on the parabola is, for example,

$$(x - a,\ y - a^2).$$

We have

$$(y - x^2) \subseteq (x - a,\ y - a^2).$$

Therefore the generic point of the parabola specializes to every ordinary closed point of the parabola.

7.6 The Origin in $\mathbb{A}^2_k$

Let

$$Z = V(x, y) \subseteq \mathbb{A}^2_k.$$

This is the origin:

$$(0, 0).$$

The ideal

$$(x, y)$$

is maximal, hence prime. Therefore Z is irreducible.

Its generic point is

$$\eta = (x, y).$$

Since Z consists of only one point,

$$\overline{\{(x, y)\}} = \{(x, y)\} = V(x, y).$$

Hence

$$\boxed{(x, y) \text{ is the generic point of the origin.}}$$

Again, a closed point is its own generic point.

7.7 A Reducible Example: $V(xy) \subseteq \mathbb{A}^2_k$

Let

$$X = \mathbb{A}^2_k = \operatorname{Spec} k[x, y]$$

and let

$$Z = V(xy).$$

Since

$$xy = 0$$

means

$$x = 0 \qquad \text{or} \qquad y = 0,$$

we have
$$V(xy) = V(x) \cup V(y).$$

Thus Z is the union of two lines:
$$V(x) = \text{the } y\text{-axis}$$
and
$$V(y) = \text{the } x\text{-axis}.$$

This closed subset is not irreducible. Therefore Problem 1 does not give one generic point for all of Z.

Instead, each irreducible component has its own generic point:
$$\boxed{(x) \text{ is the generic point of } V(x),}$$
and
$$\boxed{(y) \text{ is the generic point of } V(y).}$$

Diagrammatically,
$$\begin{array}{ccc} & V(xy) & \\ \swarrow & & \searrow \\ V(x) & & V(y) \\ \eta = (x) & & \eta = (y) \end{array}$$

So for the reducible closed set
$$V(xy) = V(x) \cup V(y),$$
there is no single generic point of the whole set.

7.8 The Scheme $\operatorname{Spec}\mathbb{Z}$

Let
$$X = \operatorname{Spec}\mathbb{Z}.$$

The prime ideals of $\mathbb{Z}$ are
$$(0)$$
and
$$(p),$$
where p is a prime number.

The closure of (0) is
$$\overline{\{(0)\}} = V(0) = \operatorname{Spec}\mathbb{Z}.$$

Therefore
$$\boxed{(0) \text{ is the generic point of } \operatorname{Spec}\mathbb{Z}.}$$

Each point (p) is closed because

$$\mathbb{Z}/(p)$$

is a field. Hence

$$\overline{\{(p)\}} = \{(p)\}.$$

Therefore

$$\boxed{(p) \text{ is the generic point of the closed point } V(p).}$$

The specialization picture is

$$\begin{array}{ccccc} & (0) & & & \text{generic point} \\ \swarrow & \downarrow & \searrow & & \\ (2) & (3) & (5) & \cdots & \text{closed points} \end{array}$$

Thus $\operatorname{Spec} \mathbb{Z}$ behaves like an arithmetic curve: it has one generic point (0), and many closed points (p), one for each prime number.

7.9 Contrast with Ordinary Hausdorff Spaces

In an ordinary Hausdorff space, such as

$$\mathbb{R}$$

with the usual topology, every point is closed:

$$\overline{\{x\}} = \{x\}.$$

Therefore a closed subset such as

$$[0,1] \subseteq \mathbb{R}$$

does not have a generic point. No single point $x \in [0,1]$ satisfies

$$\overline{\{x\}} = [0,1].$$

Hence, in ordinary Hausdorff spaces, only one-point irreducible closed sets have generic points.

This shows why schemes are different from classical spaces. Schemes are usually not Hausdorff, and this non-Hausdorff behavior is exactly what allows irreducible closed subsets to have generic points.

7.10 Summary Table

Space	Closed irreducible subset Z	Generic point η
$\operatorname{Spec} k[t]$	$\mathbb{A}^1_k$	(0)
$\operatorname{Spec} k[t]$	$V(t-a)$	$(t-a)$
$\operatorname{Spec} k[x,y]$	$\mathbb{A}^2_k$	(0)
$\operatorname{Spec} k[x,y]$	$V(y)$	(y)
$\operatorname{Spec} k[x,y]$	$V(y-x^2)$	$(y-x^2)$
$\operatorname{Spec} k[x,y]$	$V(x,y)$	(x,y)
$\operatorname{Spec} k[x,y]$	$V(x)$	(x)
$\operatorname{Spec} k[x,y]$	$V(y)$	(y)
$\operatorname{Spec} \mathbb{Z}$	$\operatorname{Spec} \mathbb{Z}$	(0)
$\operatorname{Spec} \mathbb{Z}$	$V(p)$	(p)

The main rule is:

$$\boxed{\text{If } Z = V(\mathfrak{p}) \subseteq \operatorname{Spec} A \text{ with } \mathfrak{p} \text{ prime, then } \mathfrak{p} \text{ is the generic point of } Z.}$$

8 Generic Points: Definition, Theory, and Examples

8.1 Definition

Let X be a topological space, and let $Z \subseteq X$. A point

$$\eta \in Z$$

is called a **generic point** of Z if

$$\overline{\{\eta\}} = Z.$$

Thus a generic point is a point whose closure is the whole subset.

Equivalently,

$$\boxed{\eta \text{ is generic for } Z \iff \text{every point of } Z \text{ is a specialization of } \eta.}$$

Usually, we apply this definition when Z is a **closed irreducible subset** of a scheme.

8.2 First Consequences

If

$$Z = \overline{\{\eta\}},$$

then Z is automatically closed, because it is a closure.

Also, Z is irreducible. Indeed, the closure of a single point is always irreducible.

Therefore:

$$\boxed{\text{Only closed irreducible subsets can have generic points.}}$$

If Z is reducible, then Z cannot have one single generic point. It may have several irreducible components, and each irreducible component has its own generic point.

For example,

$$V(xy) = V(x) \cup V(y) \subseteq \mathbb{A}^2_k$$

is reducible. Hence $V(xy)$ has no single generic point. However, $V(x)$ has generic point (x), and $V(y)$ has generic point (y).

8.3 Generic Points in Schemes

The crucial theorem for schemes is:

$$\boxed{\text{Every closed irreducible subset of a scheme has a unique generic point.}}$$

This property is special. It is not true in ordinary Hausdorff spaces such as $\mathbb{R}$ with its usual topology.

For affine schemes, the statement is very concrete. Let

$$X = \operatorname{Spec} A.$$

The points of X are prime ideals

$$\mathfrak{p} \subseteq A.$$

For a point $\mathfrak{p} \in \operatorname{Spec} A$, its closure is

$$\boxed{\overline{\{\mathfrak{p}\}} = V(\mathfrak{p}).}$$

Therefore:

$$\boxed{\mathfrak{p} \text{ is the generic point of } V(\mathfrak{p}).}$$

This is the most important rule.

8.4 Specialization and Prime Ideals

Let

$$X = \operatorname{Spec} A.$$

Suppose $x = \mathfrak{p}$ and $y = \mathfrak{q}$ are two points of X, where $\mathfrak{p}, \mathfrak{q} \subseteq A$ are prime ideals.

Then

$$y \in \overline{\{x\}}$$

if and only if

$$\mathfrak{p} \subseteq \mathfrak{q}.$$

Hence

$$\boxed{\mathfrak{p} \subseteq \mathfrak{q} \iff \mathfrak{q} \text{ is a specialization of } \mathfrak{p}.}$$

Thus smaller prime ideals correspond to more generic points, while larger prime ideals correspond to more special points.

In schemes, inclusion of ideals reverses inclusion of closed subsets:

$$\mathfrak{p} \subseteq \mathfrak{q} \quad \Longrightarrow \quad V(\mathfrak{q}) \subseteq V(\mathfrak{p}).$$

8.5 Example: $\operatorname{Spec} k$

Let k be a field. Then

$$\operatorname{Spec} k = \{(0)\}.$$

There is only one point. Its closure is itself:

$$\overline{\{(0)\}} = \{(0)\}.$$

Therefore:

$$\boxed{(0) \text{ is the generic point of } \operatorname{Spec} k.}$$

In this zero-dimensional example, the unique point is both closed and generic.

8.6 Example: The Affine Line $\mathbb{A}^1_k$

Let

$$X = \mathbb{A}^1_k = \operatorname{Spec} k[t].$$

If k is algebraically closed, then the prime ideals of $k[t]$ are

$$(0)$$

and

$$(t-a), \qquad a \in k.$$

The closure of (0) is

$$\overline{\{(0)\}} = V(0) = \operatorname{Spec} k[t].$$

Therefore:

$$\boxed{(0) \text{ is the generic point of } \mathbb{A}^1_k.}$$

Each maximal ideal $(t-a)$ is a closed point:

$$\overline{\{(t-a)\}} = V(t-a) = \{(t-a)\}.$$

Thus:

$$\boxed{(t-a) \text{ is the generic point of the closed point } V(t-a).}$$

The specialization picture is:

$$\begin{array}{cccccc} & (0) & & & \text{generic point} \\ \swarrow & \downarrow & \searrow & & \\ (t-a_1) & (t-a_2) & (t-a_3) & \cdots & \text{closed points} \end{array}$$

8.7 Example: The Affine Line over $\mathbb{R}$

Let

$$X = \operatorname{Spec} \mathbb{R}[t].$$

The generic point of X is still

$$(0).$$

Indeed,

$$\overline{\{(0)\}} = V(0) = \operatorname{Spec} \mathbb{R}[t].$$

Therefore:

$$\boxed{(0) \text{ is the generic point of } \operatorname{Spec} \mathbb{R}[t].}$$

But the closed points are not only ideals of the form $(t-a)$. There are also closed points corresponding to irreducible quadratic polynomials, for example

$$(t^2+1).$$

Indeed,

$$\mathbb{R}[t]/(t^2+1) \cong \mathbb{C},$$

which is a field. Hence (t^2+1) is maximal, and therefore closed.

So:

$$\boxed{(t^2+1) \text{ is a closed point, hence generic for itself.}}$$

8.8 Example: The Affine Plane $\mathbb{A}^2_k$

Let

$$X = \mathbb{A}^2_k = \operatorname{Spec} k[x,y].$$

Since $k[x,y]$ is a domain, X is irreducible. Its generic point is

$$(0).$$

Indeed,

$$\overline{\{(0)\}} = V(0) = \operatorname{Spec} k[x,y].$$

Thus:

$$\boxed{(0) \text{ is the generic point of } \mathbb{A}^2_k.}$$

A typical specialization chain is

$$(0) \subsetneq (y) \subsetneq (x-a, y).$$

Geometrically:

Prime ideal	**Closure**	**Meaning**
(0)	$\mathbb{A}^2_k$	generic point of the plane
(y)	$V(y)$	generic point of the x-axis
$(x-a,y)$	$\{(a,0)\}$	closed point

8.9 Example: A Line in $\mathbb{A}^2_k$

Let

$$X = \mathbb{A}^2_k = \operatorname{Spec} k[x, y].$$

Consider the closed subset

$$Z = V(y).$$

This is the x-axis.

The ideal (y) is prime because

$$k[x, y]/(y) \cong k[x],$$

which is a domain.

Therefore Z is irreducible. Its generic point is

$$(y).$$

Indeed,

$$\overline{\{(y)\}} = V(y).$$

Therefore:

$$\boxed{(y) \text{ is the generic point of the } x\text{-axis.}}$$

Closed points on this line have the form

$$(x - a, y).$$

Since

$$(y) \subseteq (x - a, y),$$

every closed point on the line is a specialization of the generic point (y).

8.10 Example: A Vertical Line in $\mathbb{A}^2_k$

Let

$$Z = V(x - a) \subseteq \mathbb{A}^2_k.$$

This is the vertical line

$$x = a.$$

The ideal $(x - a)$ is prime because

$$k[x, y]/(x - a) \cong k[y],$$

which is a domain.

Therefore Z is irreducible, and its generic point is

$$(x - a).$$

Thus:

$$\boxed{(x-a) \text{ is the generic point of the vertical line } x=a.}$$

A closed point on this line has the form

$$(x-a, y-b).$$

We have

$$(x-a) \subseteq (x-a, y-b).$$

8.11 Example: A Parabola in $\mathbb{A}^2_k$

Let

$$Z = V(y-x^2) \subseteq \mathbb{A}^2_k.$$

This is the parabola

$$y = x^2.$$

The ideal

$$(y-x^2)$$

is prime because

$$k[x,y]/(y-x^2) \cong k[x],$$

which is a domain.

Therefore Z is closed and irreducible. Its generic point is

$$(y-x^2).$$

Indeed:

$$\overline{\{(y-x^2)\}} = V(y-x^2).$$

Hence:

$$\boxed{(y-x^2) \text{ is the generic point of the parabola } y=x^2.}$$

A closed point on the parabola is

$$(x-a, y-a^2).$$

Since

$$(y-x^2) \subseteq (x-a, y-a^2),$$

the generic point of the parabola specializes to every ordinary point of the parabola.

8.12 Example: A Circle over $\mathbb{R}$

Let
$$Z = V(x^2 + y^2 - 1) \subseteq \operatorname{Spec} \mathbb{R}[x, y].$$

The polynomial
$$x^2 + y^2 - 1$$
is irreducible over $\mathbb{R}$. Hence the ideal
$$(x^2 + y^2 - 1)$$
is prime in $\mathbb{R}[x, y]$.

Therefore Z is irreducible.

Its generic point is
$$(x^2 + y^2 - 1).$$

Thus:
$$\boxed{(x^2 + y^2 - 1) \text{ is the generic point of the circle.}}$$

This generic point is not an ordinary real point of the circle. It represents the whole irreducible algebraic curve.

8.13 Example: The Origin in $\mathbb{A}_k^2$

Let
$$Z = V(x, y) \subseteq \mathbb{A}_k^2.$$

This is the origin $(0, 0)$.

The ideal (x, y) is maximal, hence prime. Therefore Z is irreducible.

Since Z consists of one point, its generic point is itself:
$$\eta = (x, y).$$

Indeed:
$$\overline{\{(x, y)\}} = V(x, y) = \{(x, y)\}.$$

Thus:
$$\boxed{(x, y) \text{ is the generic point of the origin.}}$$

8.14 Example: Whole Affine Space $\mathbb{A}_k^n$

Let
$$X = \mathbb{A}_k^n = \operatorname{Spec} k[x_1, \dots, x_n].$$

Since $k[x_1, \dots, x_n]$ is a domain, X is irreducible.

Its generic point is

$$(0).$$

Therefore:

$$\boxed{(0) \text{ is the generic point of } \mathbb{A}^n_k.}$$

If k is algebraically closed, a closed point has the form

$$(x_1 - a_1, \ldots, x_n - a_n).$$

A typical specialization chain is

$$(0) \subseteq (x_1 - a_1) \subseteq (x_1 - a_1, x_2 - a_2) \subseteq \cdots \subseteq (x_1 - a_1, \ldots, x_n - a_n).$$

8.15 Example: An Irreducible Hypersurface

Let

$$Z = V(f) \subseteq \mathbb{A}^n_k,$$

where

$$f \in k[x_1, \ldots, x_n].$$

If (f) is a prime ideal, then Z is irreducible.

In that case, the generic point of Z is

$$(f).$$

Hence:

$$\boxed{(f) \text{ is the generic point of the hypersurface } V(f).}$$

For example, consider

$$Z = V(z - x^2 - y^2) \subseteq \mathbb{A}^3_k.$$

Since

$$k[x, y, z]/(z - x^2 - y^2) \cong k[x, y],$$

the quotient is a domain. Therefore $(z - x^2 - y^2)$ is prime.

So the generic point of Z is

$$(z - x^2 - y^2).$$

8.16 Example: Reducible Set $V(xy)$

Let

$$Z = V(xy) \subseteq \mathbb{A}^2_k.$$

Since

$$xy = 0$$

means

$$x = 0 \qquad \text{or} \qquad y = 0,$$

we have
$$V(xy) = V(x) \cup V(y).$$

Thus Z is reducible.

Therefore Z has no single generic point.

But its irreducible components do have generic points:

$$\boxed{(x) \text{ is the generic point of } V(x),}$$

and

$$\boxed{(y) \text{ is the generic point of } V(y).}$$

So:

$$\boxed{V(xy) \text{ has two generic points for its two components, but no generic point for the whole set.}}$$

8.17 Example: A Nonreduced Scheme

Let
$$A = k[\varepsilon]/(\varepsilon^2),$$
and let
$$X = \operatorname{Spec} A.$$

The only prime ideal of A is
$$(\varepsilon).$$

Therefore X has only one point. Its closure is the whole space:

$$\overline{\{(\varepsilon)\}} = X.$$

Thus:
$$\boxed{(\varepsilon) \text{ is the generic point of } \operatorname{Spec} k[\varepsilon]/(\varepsilon^2).}$$

This example is important because X is not reduced, but it still has a generic point. Generic points are topological; they do not require the scheme to be reduced.

8.18 Example: $\operatorname{Spec} \mathbb{Z}$

Let
$$X = \operatorname{Spec} \mathbb{Z}.$$

The prime ideals are
$$(0)$$
and
$$(p),$$
where p is a prime number.

The closure of (0) is

$$\overline{\{(0)\}} = V(0) = \operatorname{Spec}\mathbb{Z}.$$

Therefore:

$$\boxed{(0) \text{ is the generic point of } \operatorname{Spec}\mathbb{Z}.}$$

Each (p) is closed because

$$\mathbb{Z}/(p)$$

is a field.

Thus:

$$\boxed{(p) \text{ is the generic point of the closed point } V(p).}$$

The specialization picture is:

$$\begin{array}{ccccl} & (0) & & & \text{generic point} \\ \swarrow & \downarrow & \searrow & & \\ (2) & (3) & (5) & \cdots & \text{closed points} \end{array}$$

This is why $\operatorname{Spec}\mathbb{Z}$ behaves like an arithmetic curve.

8.19 Example: Ordinary Hausdorff Space $\mathbb{R}$

Let

$$X = \mathbb{R}$$

with the usual topology.

Every point is closed:

$$\overline{\{x\}} = \{x\}.$$

Therefore an interval such as

$$[0, 1]$$

does not have a generic point.

Indeed, there is no point $\eta \in [0, 1]$ such that

$$\overline{\{\eta\}} = [0, 1].$$

Hence:

$$\boxed{[0, 1] \subseteq \mathbb{R} \text{ has no generic point.}}$$

This shows that schemes behave very differently from ordinary Hausdorff spaces.

8.20 Example: The Sierpiński Space

The Sierpiński space has two points:

$$X = \{\eta, p\}.$$

Its open sets are

$$\varnothing, \qquad \{\eta\}, \qquad X.$$

Hence the closed sets are

$$\varnothing, \qquad \{p\}, \qquad X.$$

The closure of η is

$$\overline{\{\eta\}} = X.$$

The closure of p is

$$\overline{\{p\}} = \{p\}.$$

Therefore:

$$\boxed{\eta \text{ is the generic point of } X,}$$

and

$$\boxed{p \text{ is a closed point.}}$$

This is the simplest topological model of a scheme-like specialization relation.

8.21 Summary Table

Space	**Closed irreducible subset Z**	**Generic point**
$\operatorname{Spec} k$	$\operatorname{Spec} k$	(0)
$\operatorname{Spec} k[t]$	$\mathbb{A}^1_k$	(0)
$\operatorname{Spec} k[t]$	$V(t-a)$	$(t-a)$
$\operatorname{Spec} \mathbb{R}[t]$	$\operatorname{Spec} \mathbb{R}[t]$	(0)
$\operatorname{Spec} \mathbb{R}[t]$	$V(t^2+1)$	(t^2+1)
$\operatorname{Spec} k[x,y]$	$\mathbb{A}^2_k$	(0)
$\operatorname{Spec} k[x,y]$	$V(y)$	(y)
$\operatorname{Spec} k[x,y]$	$V(x-a)$	$(x-a)$
$\operatorname{Spec} k[x,y]$	$V(y-x^2)$	$(y-x^2)$
$\operatorname{Spec} k[x,y]$	$V(x,y)$	(x,y)
$\operatorname{Spec} k[x_1,\ldots,x_n]$	$\mathbb{A}^n_k$	(0)
$\operatorname{Spec} k[x,y,z]$	$V(z-x^2-y^2)$	$(z-x^2-y^2)$
$\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$	$\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$	(ε)
$\operatorname{Spec} \mathbb{Z}$	$\operatorname{Spec} \mathbb{Z}$	(0)
$\operatorname{Spec} \mathbb{Z}$	$V(p)$	(p)
$\mathbb{R}$ usual topology	$[0,1]$	none
Sierpiński space	X	η

8.22 Main Rule

The main rule to remember is:

$$\boxed{\text{In } \operatorname{Spec} A, \text{ if } Z = V(\mathfrak{p}) \text{ with } \mathfrak{p} \text{ prime, then } \mathfrak{p} \text{ is the generic point of } Z.}$$

Equivalently,

$$\boxed{\overline{\{\mathfrak{p}\}} = V(\mathfrak{p}).}$$